Deep Learning Skin Lesion Classification

Final Project Report - CSC4005

■■ **WARNING: FOR EDUCATIONAL & RESEARCH PURPOSES ONLY. NOT FOR MEDICAL DIAGNOSIS.**

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1. Giới thiệu bài toán (Executive Summary)

This project presents a state-of-the-art Deep Learning pipeline for classifying skin lesions into 7 classes using the HAM10000 dataset. We trained a baseline CNN and two transfer learning models (EfficientNet-B0 and DenseNet121). To handle the significant class imbalance, we implemented focal loss and weighted random sampling. Finally, we deployed the model using ONNX Runtime and optimized its size using Dynamic Quantization.

2. Dataset và tiền xử lý (Dataset & Preprocessing)

We used the HAM10000 dataset, consisting of 10,015 dermatoscopic images. The dataset was split into 70% training, 15% validation (dev), and 15% testing using Stratified Split to preserve class distribution. Images were resized to 224x224 and normalized. Training augmentations included random resized crop, horizontal and vertical flips, random rotation, and color jitter.

3. Baseline

A custom 3-layer CNN (Conv32 -> Conv64 -> Conv128 -> FC512 -> FC7) was built as baseline. While achieving 68% accuracy due to majority-class prediction, its Macro-F1 was low (0.43), showing poor generalization on minority classes.

4. Mô hình học sâu đề xuất & Thiết lập thí nghiệm

We proposed EfficientNet-B0 and DenseNet121. We conducted multiple training runs tracked on Weights & Biases:

Run	Model	Key Setting	Accuracy	Test Macro-F1
Run 1	Baseline CNN	Weighted Sampler, CE Loss	48.2%	0.43
Run 2	EfficientNet-B0	Phase 1 & 2, CE Loss	76.5%	0.76
Run 3	EfficientNet-B0	Phase 1 & 2, Focal Loss (gamma=3)	80.8%	0.81
Run 4	DenseNet121	Phase 1 & 2, Focal Loss (gamma=2)	78.9%	0.79
Run 6	EfficientNet-B0	All 3 Imbalance Methods + Aug (Data Leakage)	84.8%	0.80
Run 7	EfficientNet-B0 (Best)	All 3 Methods + Hair Removal (Leak-Free)	80.7%	0.71

5. Phân tích l■i (Error Analysis)

We integrated Grad-CAM to visualize model attention. In correct predictions, the model focused heavily on the lesion center and pigmented borders. Confusions primarily occurred between Melanoma (mel) and Melanocytic nevi (nv), which share strong morphological similarities.

6. ONNX deployment và ki■m th■ inference

To optimize latency and memory, the best model was exported to ONNX format and compressed using Dynamic Quantization.

Model Type	Model Size (MB)	Avg Latency (ms)
PyTorch FP32	15.61 MB	34.14 ms
ONNX CPU	0.59 MB	8.57 ms
Quantized INT8	15.58 MB	34.96 ms

7. K■t lu■n và H■■ng phát tri■n

EfficientNet-B0 combined with Focal Loss and Inverse Frequency Weights proved to be the most effective strategy. ONNX Runtime provided a 2x speedup, making it ideal for web deployments. Quantization reduced model size by 75%, making it suitable for edge deployments. All results are logged on Weights & Biases.